Summer Math Adventure

W2 D1

Week 2 · Day 1 | Fractions: Meaning & Equivalent Fractions

Name: ______________________ Date: ______________________

FRACTIONS

Fraction = part of a whole. $\dfrac{\textbf{numerator}}{\textbf{denominator}}$ → numerator = how many parts you have, denominator = total equal parts.

Equivalent fractions: multiply OR divide numerator AND denominator by the same number.

$$\frac{2}{3} = \frac{2\times4}{3\times4} = \frac{8}{12}$$

Part A – Shade & Label

Shade the correct fraction of each bar. Then write an equivalent fraction by doubling.

1. Shade $\frac{3}{4}$ — then find an equivalent fraction with denominator 8.

Equivalent: $\frac{3}{4} = \frac{\square}{8}$

2. Shade $\frac{2}{5}$ — then find an equivalent fraction with denominator 15.

Equivalent: $\frac{2}{5} = \frac{\square}{15}$

Part B – Find the Missing Number

3. $\frac{3}{5} = \frac{\square}{20}$

4. $\frac{4}{6} = \frac{2}{\square}$

5. $\frac{5}{8} = \frac{\square}{40}$

6. $\frac{18}{24} = \frac{3}{\square}$

Part C – Simplest Form

Divide numerator and denominator by their GCF (Greatest Common Factor).

7. $\frac{8}{12} = \frac{\square}{\square}$

8. $\frac{15}{20} = \frac{\square}{\square}$

9. $\frac{24}{36} = \frac{\square}{\square}$

10. $\frac{30}{45} = \frac{\square}{\square}$

Part D – Compare Fractions

Write <, >, or = . First convert to a common denominator.

11. $\frac{3}{4} \; \square \; \frac{5}{8}$

Common denom: ______________________

12. $\frac{2}{3} \; \square \; \frac{3}{5}$

Common denom: ______________________

Summer Math Adventure

W2 D2

Week 2 · Day 2 | Adding & Subtracting Fractions with Unlike Denominators

Name: ______________________ Date: ______________________

FRACTIONS: ADD & SUBTRACT

To add or subtract fractions with different denominators:

Step 1: Find the **LCD** (Least Common Denominator)

Step 2: Convert each fraction to an equivalent fraction using the LCD

Step 3: Add or subtract the numerators. Keep the denominator.

Step 4: Simplify if needed.

Example: $\frac{1}{3} + \frac{1}{4}$ → LCD = 12 → $\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$

Part A – Find the LCD First

1. Denominators: 4 and 6 LCD = ☐

2. Denominators: 3 and 8 LCD = ☐

3. Denominators: 5 and 6 LCD = ☐

4. Denominators: 4 and 9 LCD = ☐

Part B – Add the Fractions

Show the LCD step. Simplify your answer.

5. $\frac{1}{2} + \frac{1}{5} =$

LCD: ☐

$\frac{☐}{☐} + \frac{☐}{☐} =$

____

6. $\frac{2}{3} + \frac{1}{4} =$

LCD: ☐

$\frac{☐}{☐} + \frac{☐}{☐} =$

____

7. $\frac{3}{4} + \frac{2}{5} =$

LCD: ☐

=

____

8. $\frac{5}{6} + \frac{3}{8} =$

LCD: ☐

=

____

Part C – Subtract the Fractions

9. $\frac{3}{4} - \frac{1}{3} =$

____

10. $\frac{5}{6} - \frac{1}{4} =$

____

Summer Math Adventure

W2 D3

Week 2 · Day 3 | Mixed Numbers: Converting, Adding & Subtracting

Name: ____________________ Date: ____________________

MIXED NUMBERS

Mixed Number: whole number + fraction (e.g., $3\frac{1}{2}$)

Convert to improper fraction: multiply whole × denominator, add numerator, keep denominator.

$3\frac{1}{2} \rightarrow (3\times2)+1 = 7 \rightarrow \frac{7}{2}$

Convert to mixed number: divide. Quotient = whole, remainder = new numerator.

$\frac{11}{4} \rightarrow 11\div4 = 2 \text{ R } 3 \rightarrow 2\frac{3}{4}$

Part A – Convert Between Forms

Mixed → Improper:

1. $2\frac{3}{4} = \frac{\square}{\square}$
2. $5\frac{2}{3} = \frac{\square}{\square}$
3. $4\frac{5}{6} = \frac{\square}{\square}$

Improper → Mixed:

4. $\frac{17}{5} =$ ____________________
5. $\frac{29}{4} =$ ____________________
6. $\frac{41}{6} =$ ____________________

Part B – Add Mixed Numbers

Add whole numbers + add fractions. Convert to LCD first. Regroup if fraction ≥ 1.

7. $3\frac{1}{2} + 2\frac{1}{4} =$
8. $4\frac{2}{3} + 1\frac{3}{4} =$
9. $2\frac{5}{6} + 3\frac{3}{4} =$
10. $6\frac{7}{8} + 2\frac{3}{4} =$

Part C – Subtract Mixed Numbers

Summer Math Adventure

W2 D4

Week 2 · Day 4 | Multiplying Fractions & Fractions of Whole Numbers

Name: ______________________ Date: ______________________

MULTIPLYING FRACTIONS

Multiply fractions: top × top, bottom × bottom.

$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$

Fraction of a whole number: multiply the whole number by the fraction.

$\frac{3}{4}$ of $24 = \frac{3}{4} \times 24 = \frac{3 \times 24}{4} = \frac{72}{4} = 18$

Tip: Cross-cancel before multiplying to keep numbers small!

Part A – Multiply Fractions

1. $\frac{1}{2} \times \frac{3}{5} =$ ______________________

2. $\frac{3}{4} \times \frac{2}{9} =$ ______________________

3. $\frac{5}{6} \times \frac{3}{10} =$ ______________________

4. $\frac{4}{7} \times \frac{7}{8} =$ ______________________

5. $\frac{2}{3} \times \frac{6}{7} =$ ______________________

6. $\frac{5}{8} \times \frac{4}{5} =$ ______________________

Part B – Fraction of a Whole Number

7. $\frac{3}{4}$ of 32 = ______________________

8. $\frac{2}{5}$ of 45 = ______________________

9. $\frac{5}{6}$ of 54 = ______________________

10. $\frac{7}{8}$ of 64 = ______________________

Part C – Word Problems

11. A hiking trail is $4\frac{1}{2}$ miles. Sofia walked $\frac{2}{3}$ of the trail. How far did she walk?

Answer: ______________________

12. A class of 30 students: $\frac{3}{5}$ chose soccer, $\frac{1}{6}$ chose swimming, and the rest chose basketball. How many chose each sport?

Summer Math Adventure

W2
D5

Week 2 · Day 5 | Fractions on Number Lines & Cumulative Review

Name: ____________________ Date: ____________________

FRACTIONS: REVIEW

Number line tip: The space between 0 and 1 is divided into equal parts equal to the denominator. Count "jumps" equal to the numerator.

Part A – Number Line Fractions

Label the marked points on each number line.

1. Each section = $\frac{1}{6}$. Label A, B, C.

A = __________ B = __________ C = __________

2. Plot these fractions on the number line: $\frac{1}{4}$, $\frac{3}{4}$, $\frac{1}{2}$

Part B – Mixed Review Sprint

3. $\frac{3}{8} + \frac{1}{4} =$ ____________________

4. $\frac{5}{6} - \frac{1}{3} =$ ____________________

5. $\frac{3}{5} \times \frac{2}{3} =$ ____________________

6. $3\frac{1}{2} + 2\frac{3}{4} =$ ____________________

7. $5\frac{1}{3} - 1\frac{2}{3} =$ ____________________

8. $\frac{2}{3}$ of 27 = ____________________

Part C – Cumulative: Number Sense + Fractions

9. A bucket holds $3\frac{3}{4}$ gallons. You fill it $\frac{2}{3}$ full. How many gallons are in the bucket?

Answer: ____________________

10. A school raised $48,360. They spent $\frac{3}{8}$ on supplies. How much was spent on supplies? (Hint: find

3

Week 2 Quiz – Fractions

W2 QUIZ

Topics: Equivalence · Comparing · Add/Subtract · Mixed Numbers · Multiply

Name: ______________________ Date: ______________________

Score: _______ / 20

Section 1 – Equivalence & Comparing (5 pts)

1. Fill in the blank: $\frac{3}{4} = \frac{\square}{28}$

2. Simplify: $\frac{20}{28} = \frac{\square}{\square}$

3. Compare: $\frac{5}{8} \ \square \ \frac{3}{5}$

4. Order least to greatest: $\frac{2}{3}, \frac{3}{4}, \frac{5}{8}$

____

5. **(1 pt)** Convert $6\frac{3}{5}$ to an improper fraction.

____

Section 2 – Add & Subtract (7 pts)

6. $\frac{1}{3} + \frac{1}{6} =$

____

7. $\frac{5}{6} - \frac{3}{8} =$

____

8. $4\frac{3}{4} + 2\frac{2}{3} =$

____

9. $7\frac{1}{4} - 3\frac{3}{4} =$

____

10. **(3 pts)** Lena has $5\frac{1}{2}$ feet of ribbon. She uses $2\frac{3}{8}$ feet for one gift and $1\frac{3}{4}$ feet for another. How much ribbon remains?

Answer:

Section 3 – Multiply (5 pts + bonus)